

1. Diagnostic Benchmark



Restored Input
PSNR 15.9dB



Degraded Input
PSNR 7.4dB

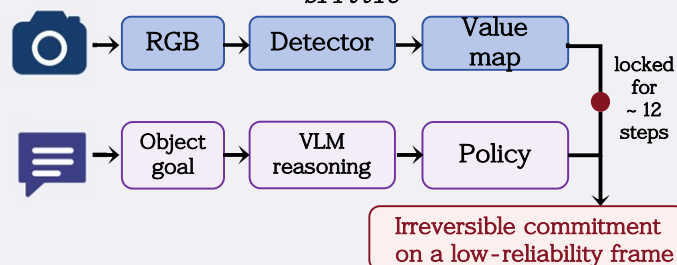
Image quality ↑
7.4 → 15.9 dB

Success rate
37% → 41%

Restoring the image does not restore navigation

2. Commitment bottleneck

Preception improves, but commitment remains brittle



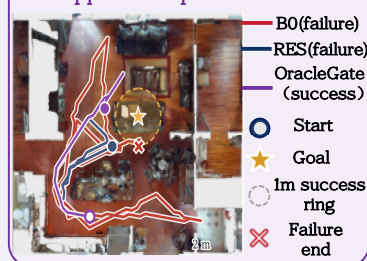
commit reliability: 0.39

Reach 61%

Success 37%

Reach-to-Success Gap:
24 pts

Appendix top-down view



The agent often reaches the goal neighborhood but fails to convert reach into success

3. Test-time patches fail to recover

Across tested intervention families, deployable gains are ~0 negative

- Pixel restoration
- Oracle-gated appearance upper bound
- Reliability-gated reweighting
- Commit-timing recovation
- Forced commit
- Temporal / multi-view aggregation

Deployable test-time interventions: ~0.04 ΔSR or negative

Degradation-aware training

Learned multi-modal sensing

Implication: retraining and learned sensing, rather than test-time patching.